

31/12/2025

Work done

- **Python tkinter**

Description

Python tkinter

- Tkinter is python's standard GUI library
- It used to create windows , labels , forms , dialogs etc.
- Tkinter = python module
- GUI = graphical user interface

Tkinter Main Components (Widgets)

Common Widgets

Widget	Purpose
Label	Display text
Button	Clickable button
Entry	Input field
Text	Multi-line text
Checkbutton	Checkbox
Radiobutton	Radio option
Listbox	List items
Frame	Container
Canvas	Drawing

Importing tkinter

import tkinter as tk

Or

from tkinter import *

Creating the main window

- Every tkinter program starts with main window
- Tk() starts Tkinter system - It creates main application window

```
import tkinter as tk

window = tk.Tk()
```

Window properties

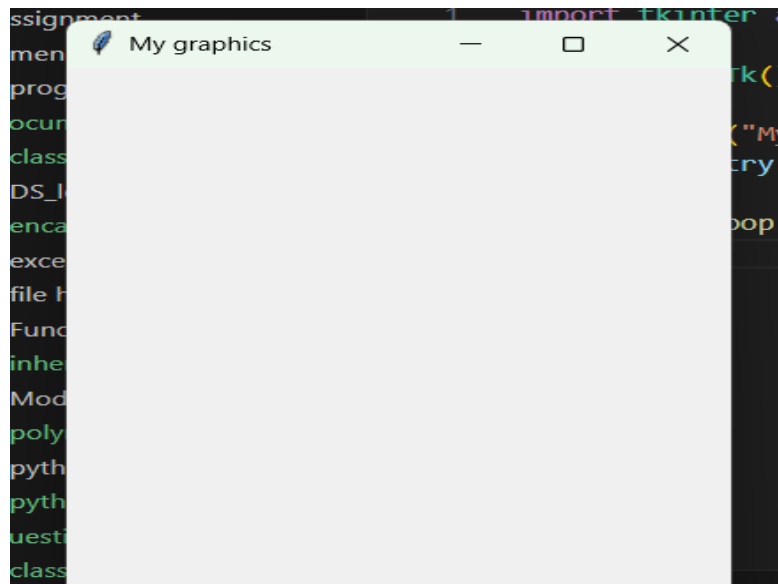
```
import tkinter as tk

window = tk.Tk()

window.title("My graphics") # title
window.geometry("300x400") # size

window.mainloop()
```

Output:



Adding label in frame

```
import tkinter as tk

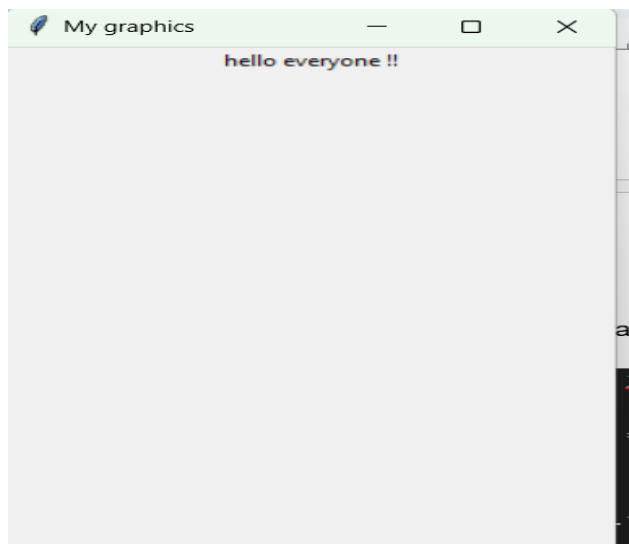
window = tk.Tk()

window.title("My graphics") # title
window.geometry("300x400") # size

label = tk.Label(window , text="hello everyone !!") # adding label widget
label.pack()

window.mainloop()
```

Output:



Adding button

```
import tkinter as tk

window = tk.Tk()

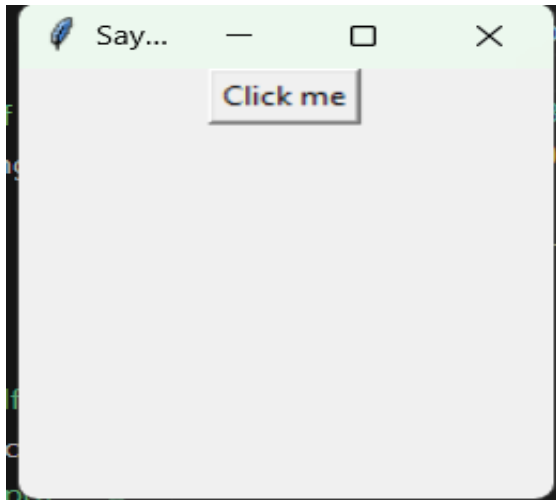
window.title("Sayma")
window.geometry("200x200")

btn = tk.Button(window, text ="Click me")
```

```
btn.pack()

window.mainloop()
```

Output:



Adding button using function

```
import tkinter as tk

window = tk.Tk()

window.title("Sayma")
window.geometry("200x200")

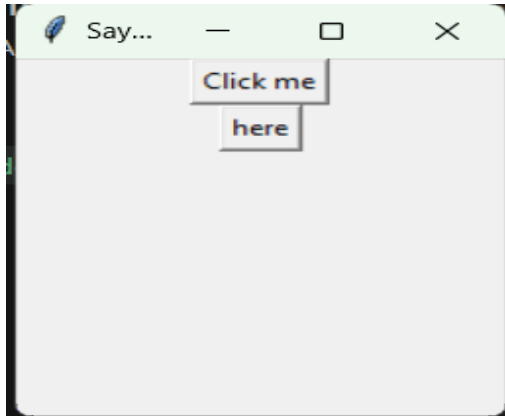
def click_me():
    print("Button clicked")

btn = tk.Button(window, text="Click me")
btn.pack()

btn1 = tk.Button(window, text="here", command=click_me)
btn1.pack()

window.mainloop()
```

Output:



After clicking button here message will print on terminal

Button clicked

Adding button and label and change the label value

```
import tkinter as tk

window = tk.Tk()

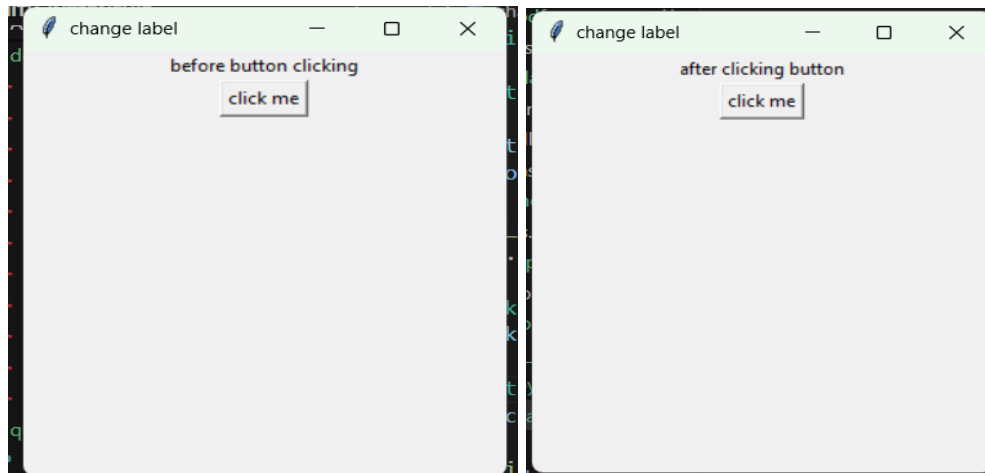
window.title("change label")
window.geometry("300x300")

def click_me():
    label.config(text="after clicking button")

label = tk.Label(window, text="before button clicking")
label.pack()

button = tk.Button(window, text="click me", command=click_me)
button.pack()

window.mainloop()
```



Adding all components and give text after button

```
import tkinter as tk
window = tk.Tk()

window.title("Entery")
window.geometry("300x300")
def show_text():
    label2.config(text="form submitted successfully...")

label = tk.Label(window, text="hello GPP student")
label.pack()

label1 = tk.Label(window, text="Enter your name")
label1.pack()

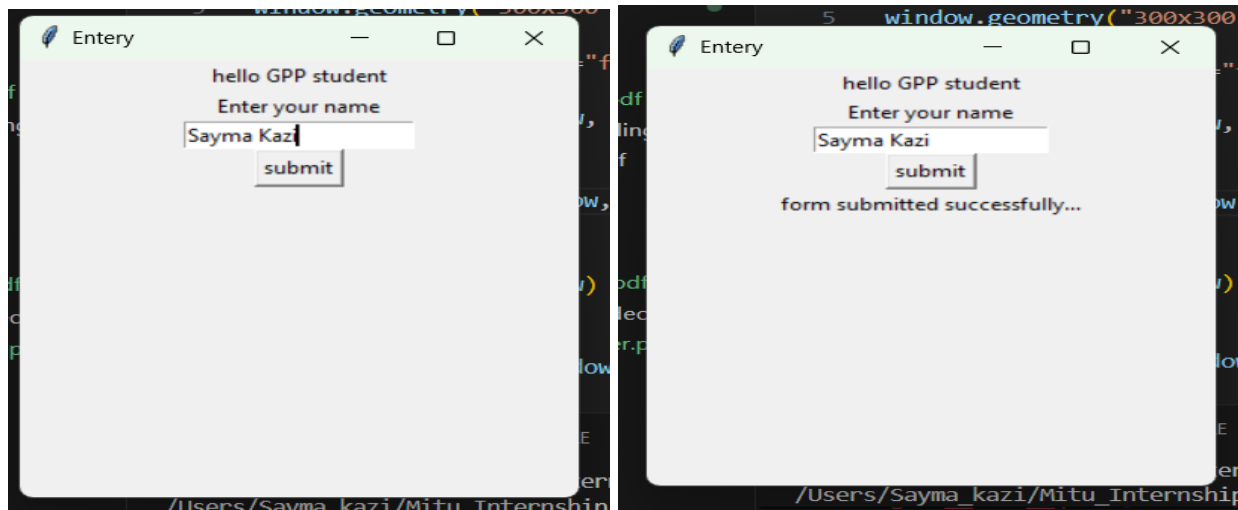
entry = tk.Entry(window)
entry.pack()

button = tk.Button(window, text="submit", command= show_text)
button.pack()

label2 = tk.Label(window , text=" ")
label2.pack()
```

```
window.mainloop()
```

Output:



Reading entries

```
import tkinter as tk

window = tk.Tk()

window.title("see this")
window.geometry("300x300")

def show2():
    print(entry.get())

l1 = tk.Label(window, text= "welcome to programming")
l1.pack()

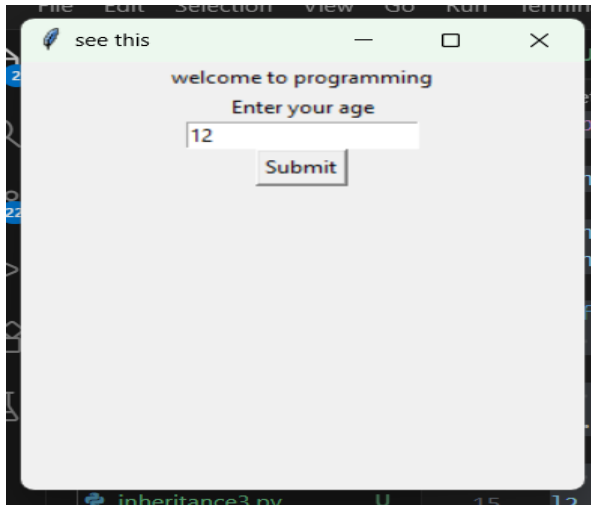
l2 = tk.Label(window, text="Enter your age")
l2.pack()

entry = tk.Entry(window)
entry.pack()

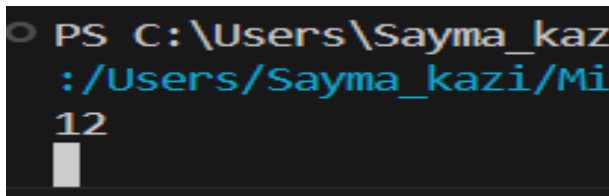
btn = tk.Button(window, text="Submit", command=show2)
btn.pack()
```

```
window.mainloop()
```

Output:



Value in entry are printed on terminal.



Grid layout

```
import tkinter as tk

wd = tk.Tk()

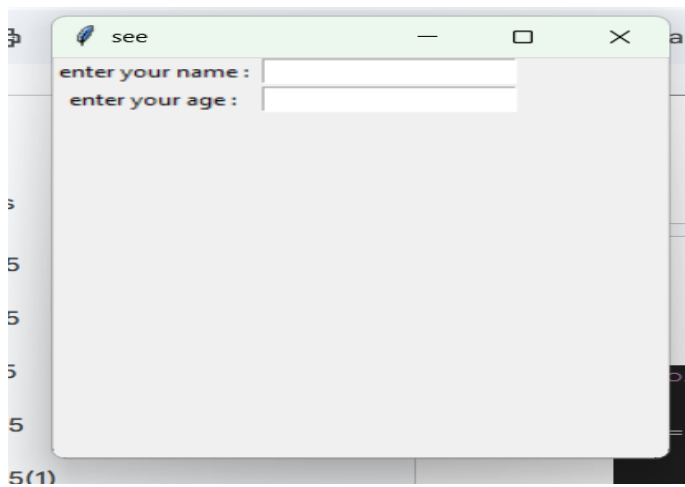
wd.title("see")
wd.geometry("300x300")

tk.Label(wd, text="enter your name : ").grid(row= 0 , column= 0)
tk.Entry(wd).grid(row=0 , column=1)

tk.Label(wd, text="enter your age : ").grid(row=1,column=0)
tk.Entry(wd).grid(row=1,column=1)

wd.mainloop()
```


Output:



Variables in Tkinter (**StringVar**, **IntVar**)

Tkinter uses **special variables**

```
name = tk.StringVar()
```

Example

```
entry = tk.Entry(window, textvariable=name)
```

```
print(name.get())
```

Checkbutton:

```
import tkinter as tk

wd = tk.Tk()

wd.title("check button")
wd.geometry("300x300")

var = tk.IntVar() # store checkbox status

def check_status():
```

```

        #var.get() - read value
    if var.get() == 1:
        # 1: checked 0: unchecked
        label.config(text="you agreed ...")
    else:
        label.config(text="you did not agree...")

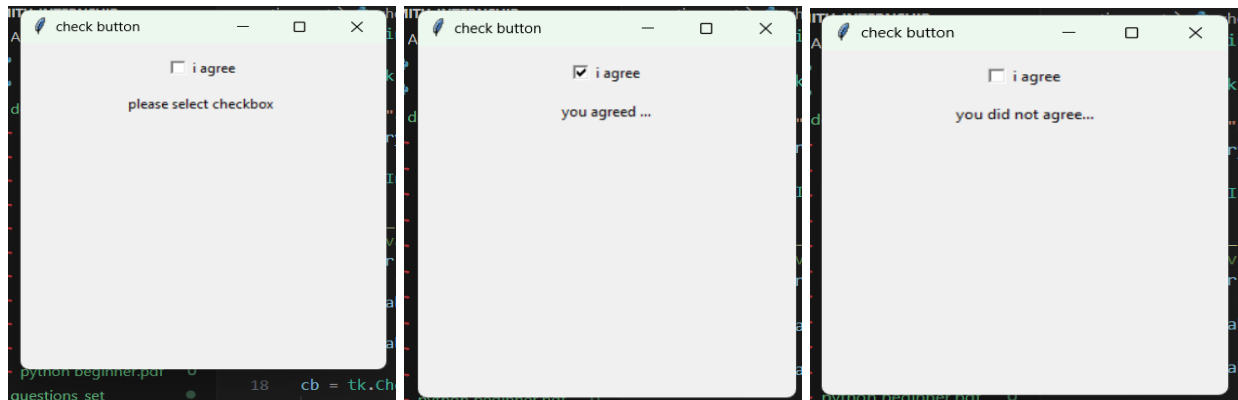
cb = tk.Checkbutton(

    wd,
    text="i agree",
    variable=var, # connect checkbox to variable
    command=check_status
)
cb.pack(pady=10)

label = tk.Label(wd, text="please select checkbox ")
label.pack()

wd.mainloop()

```



Radio button :

```

import tkinter as tk

wd = tk.Tk()

wd.geometry("300x300")

v = tk.IntVar()

```

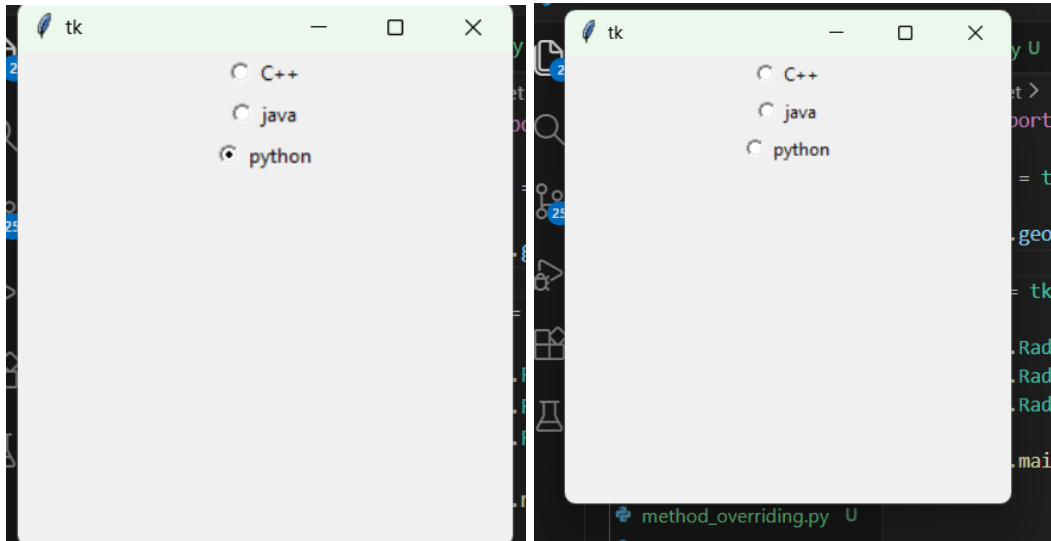
```

tk.Radiobutton(wd, text="C++", variable=v, value=1).pack()
tk.Radiobutton(wd, text="java", variable=v, value=2).pack()
tk.Radiobutton(wd, text="python", variable=v, value=3).pack()

wd.mainloop()

```

Output :



Pop up message :

```

import tkinter as tk
from tkinter import messagebox

w = tk.Tk()
w.title("MessageBox Example")

w.geometry("300x400")

def show_message():
    messagebox.showinfo("Info", "Hello User")

def warning():
    messagebox.showwarning("Warning", "This is a warning")

def error():
    messagebox.showerror("Error", "Something went wrong")

```

```
def ask():
    result = messagebox.askyesno("Confirm", "Do you want to continue?")
    if result:
        messagebox.showinfo("Result", "You clicked YES")
    else:
        messagebox.showinfo("Result", "You clicked NO")

tk.Button(w, text="Show Message", command=show_message).pack(pady=10)
tk.Button(w, text="Show Warning", command=warning).pack(pady=10)
tk.Button(w, text="Show Error", command=error).pack(pady=10)
tk.Button(w, text="Show Permission", command=ask).pack(pady=10)

w.mainloop()
```

Output :

